POLITECNICO DI MILANO
High Performance Computing Engineering
Quantum Physics – September 4, 2025

Please justify each answer and use clear handwriting. Define each symbol you employ. For each exercise, give the symbolic solution and substitute numerical values (if required) only at the end. Write your first name, family name, and personal code on each sheet in a readable way. No electronic devices are allowed.

1. Consider two non-interacting identical free particles that are constrained to move along the z axis.

a) Determine all possible wavefunctions when the particles are ½-spin fermions in a spin-triplet configuration with total momentum P_{tot}. Ensure the total wavefunction respects the appropriate symmetry under particle exchange.

b) Repeat point (a) for fermions in a spin-singlet configuration.

c) Using the results from points (a) and (b), find the probability density function for finding the two particles separated by a distance D along the z axis, for both singlet and triplet spin states.

[*Hint*: express the two-particle wavefunction in terms of the center-of-mass coordinate $z_{\text{cm}} = \frac{z_1+z_2}{2}$ and the relative distance $D = z_1 - z_2$]

d) Repeat point (c) for spinless bosons.

2. A particle of mass m is confined in the xy plane and is subject to a two-dimensional isotropic harmonic potential given by $V(x,y) = \frac{1}{2}k(x^2 + y^2)$.

a) Demonstrate that the operator $\hat{L}_z$, representing the projection of the particle's angular momentum along the z axes perpendicular to the xy plane, commutes with the system's Hamiltonian $\hat{H}$.

[*Hint*: in polar coordinates $\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}$]

b) Show that the time-independent Schrödinger equation for this system can be separated into two independent equations, one depending only on the x coordinate and the other only on the y coordinate, each corresponding to a one-dimensional quantum harmonic oscillator. Then express $\hat{H}$ in terms of the ladder (creation and annihilation) operators $\hat{a}_x^\dagger$, $\hat{a}_x$ and $\hat{a}_y^\dagger$, $\hat{a}_y$, which are associated with the one-dimensional harmonic oscillators in the x and y directions, respectively.

c) Express the operator $\hat{L}_z$ as a function of the ladder operators.

d) Find the eigenvalues of $\hat{L}_z$ for eigenstates obtained as linear combinations of the two energy-degenerate states $|\psi_1\rangle = |n_x = 1, n_y = 0\rangle$ and $|\psi_2\rangle = |n_x = 0, n_y = 1\rangle$, where n_x and n_y denote the occupation numbers of the harmonic oscillators in the x and y directions, respectively.

Solution to the 1st exercise

a) For the spin *triplet* state of two non-interacting identical *fermions*, the spatial part of the wavefunction must be *antisymmetric* under exchange of particle coordinates, as required by the Pauli exclusion principle. This is achieved by constructing the anti-symmetrized combination of single-particle plane waves:

$$\psi(z_1, z_2) = \frac{1}{L\sqrt{2}}\left(e^{ik_1 z_1} e^{ik_2 z_2} - e^{ik_1 z_2} e^{ik_2 z_1}\right) = \frac{i\sqrt{2}}{L} e^{i\frac{P_{\text{tot}}}{\hbar} z_{\text{cm}}} \sin\left(\frac{\Delta k\, D}{2}\right),$$

with $P_{\text{tot}} = \hbar(k_1 + k_2)$ and $\Delta k = k_1 - k_2$. Note that Δk can assume any value compatible with the length of the region along the z axis in which the particles are confined.

b) For the spin *singlet* state of two non-interacting identical *fermions*, the total spatial wavefunction must be *symmetric* under exchange of the particle coordinates, since the spin part is antisymmetric. The symmetrized spatial wavefunction is given by:

$$\psi(z_1, z_2) = \frac{1}{L\sqrt{2}}\left(e^{ik_1 z_1} e^{ik_2 z_2} + e^{ik_1 z_2} e^{ik_2 z_1}\right) = \frac{\sqrt{2}}{L} e^{i\frac{P_{\text{tot}}}{\hbar} z_{\text{cm}}} \cos\left(\frac{\Delta k\, D}{2}\right).$$

c) The probability density function is obtained by taking the modulus squared of the total spatial wavefunction. For the *triplet* state, this yields $|\psi(z_1, z_2)^2| = \frac{2}{L^2}\sin^2\left(\frac{\Delta k\, D}{2}\right)$, while we find $|\psi(z_1, z_2)^2| = \frac{2}{L^2}\cos^2\left(\frac{\Delta k\, D}{2}\right)$ for the *singlet* state. In both cases, the probability density depends solely on the relative distance $D = z_1 - z_2$ between the particles. Therefore, these expressions represent the probability distribution functions for finding the two particles separated by a distance D in the singlet and triplet configurations, respectively.

d) For two non-interacting identical *spinless bosons*, the total spatial wavefunction must be *symmetric* under exchange of the particle coordinates. The symmetrized spatial wavefunction has been already found at point (b) and is given by:

$$\psi(z_1, z_2) = \frac{\sqrt{2}}{L} e^{i\frac{P_{\text{tot}}}{\hbar} z_{\text{cm}}} \cos\left(\frac{\Delta k\, D}{2}\right).$$

Therefore, the probability distribution for finding the two bosons at a distance D from each other is $\frac{2}{L^2}\cos^2\left(\frac{kD}{2}\right)$.

Solution to the 2nd exercise

a) In polar coordinates the operators $\hat{H}$ and $\hat{L}_z$ assume the following expression:

$$\hat{H} = -\frac{\hbar}{2m}\nabla^2 + \frac{1}{2}k(x^2+y^2) = -\frac{\hbar}{2m}\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}\right) + \frac{1}{2}r^2;$$
$$\hat{L}_z = x\hat{p}_y - y\hat{p}_x = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) = -i\hbar\frac{\partial}{\partial \theta}.$$

From these two expressions one can see that $[\hat{H}, \hat{L}_z] = 0$.

b) $\hat{H} = \hat{H}_x + \hat{H}_Y$, with $\hat{H}_x = -\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}kx^2$; $\hat{H}_y = -\frac{\hbar}{2m}\frac{\partial^2}{\partial y^2} + \frac{1}{2}ky^2$.

The time-independent Schrödinger equation splits in the following two independent equations:

$$\hat{H}_x\psi_x(x) = E_x\psi_x(x), \qquad \hat{H}_y\psi_y(x) = E_y\psi_y(y),$$

with $\psi(x,y) = \psi_x(x)\psi_y(y)$ and $E_{\text{tot}} = E_x + E_y$.

Substituting the ladder operators in the expressions for $\hat{H}_x$ and $\hat{H}_y$, one obtains:

$$\hat{H} = \hbar\omega\left(\hat{a}_x^\dagger\hat{a}_x + \hat{a}_y^\dagger\,\hat{a}_y + 1\right), \qquad \text{where } \omega = \sqrt{\frac{k}{m}}.$$

c) $x = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}_x^\dagger + \hat{a}_x\right)$; $y = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}_y^\dagger + \hat{a}_y\right)$;

$\hat{p}_x = i\sqrt{\frac{\hbar m\omega}{2}}\left(\hat{a}_x^\dagger - \hat{a}_x\right)$; $\hat{p}_y = i\sqrt{\frac{\hbar m\omega}{2}}\left(\hat{a}_y^\dagger - \hat{a}_Y\right)$.

Since $[\hat{a}_x^\dagger, \hat{a}_y] = [\hat{a}_x, \hat{a}_y^\dagger] = [\hat{a}_x^\dagger, \hat{a}_y^\dagger] = [\hat{a}_x, \hat{a}_y] = 0$, the operator $\hat{L}_z = x\hat{p}_y - y\hat{p}_x$ assumes the following expression:

$$\hat{L}_z = x\hat{p}_y - y\hat{p}_x = \frac{1}{2}i\hbar\left[\left(\hat{a}_x^\dagger + \hat{a}_x\right)\left(\hat{a}_y^\dagger - \hat{a}_y\right) - \left(\hat{a}_y^\dagger + \hat{a}_y\right)\left(\hat{a}_x^\dagger - \hat{a}_x\right)\right] = i\hbar\left(\hat{a}_y^\dagger\hat{a}_x - \hat{a}_x^\dagger\hat{a}_y\right).$$

d) Consider the state $|\psi\rangle = \alpha|n_x = 1, n_y = 0\rangle + \beta|n_x = 0, n_y = 1\rangle = \alpha|1,0\rangle + \beta|0,1\rangle$, where α and β are complex coefficients.
We want to find the solutions of the eigenvalue equation $\hat{L}_z|\psi\rangle = m|\psi\rangle$:

$$\hat{L}_z|\psi\rangle = \hat{L}_z(\alpha|1,0\rangle + \beta|0,1\rangle) = i\hbar\,(\alpha|0,1\rangle - \beta|1,0\rangle) = m|\psi\rangle = m(\alpha|1,0\rangle + \beta|0,1\rangle).$$

The previous equation can be written in matrix form as:

$$\begin{pmatrix} 0 & -i\hbar \\ +i\hbar & 0 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = m\begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

To find the nontrivial solutions, we solve the characteristic equation:

$$\det\begin{pmatrix} m & +i\hbar \\ -i\hbar & m \end{pmatrix} = 0 \quad \Rightarrow \quad m^2 - \hbar^2 = 0 \quad \Rightarrow \quad m = \pm\hbar.$$